# 📊 Employee Performance Analysis

🎯 Practical Exam

## 🧠 Part A — Theory

📊 Mean = Average | Median = Middle | Mode = Most frequent

📏 Range = Max - Min | Variance = Spread

📊 Normal = Continuous | Poisson = Discrete

📉 Skewness = Asymmetry

🎯 Conditional Probability = P(A|B)

🔄 Independent vs Mutually Exclusive

🧠 Bayes updates probability

📉 PCA reduces dimensions

## 💻 Part B — Employee Performance Analysis

### 📊 Practical Notebook

### 🧱 Step 0: Dataset Creation

In [1]:

```
import pandas as pd
import numpy as np

np.random.seed(42)
```

```
n = 4000

df = pd.DataFrame({
    "Employee_ID": range(1, n+1),
    "Department": np.random.choice(["HR","IT","Sales","Finance"], n),
    "Age": np.random.randint(22,60,n),
    "Salary": np.random.randint(20000,100000,n),
    "Projects_Completed": np.random.randint(1,20,n),
    "Working_Hours": np.random.randint(6,12,n),
    "Performance_Score": np.random.randint(40,100,n)
})


df["Promotion_Status"] = np.where(df["Performance_Score"] > 75, "Yes",
"No")

df.head()
```

**Out [1]:**

| Employee_ID | Department | Age | Salary | Projects_Completed | Working_Hours | Performanc |
|---|---|---|---|---|---|---|
| 1 | Sales | 49 | 96958 | 5 | 10 | 44 |
| 2 | Finance | 28 | 47926 | 16 | 7 | 96 |
| 3 | HR | 56 | 33180 | 4 | 8 | 98 |
| 4 | Sales | 49 | 56931 | 17 | 7 | 48 |
| 5 | Sales | 35 | 44014 | 2 | 10 | 52 |

## 📈 Step 1: Central Tendency

**In [2]:**

```
print("Mean:", df["Salary"].mean())
print("Median:", df["Salary"].median())
print("Mode:", df["Salary"].mode()[0])
```

**Out [2]:**

```
Mean: 59944.64475
Median: 59587.5
Mode: 64135
```

## 📊 Step 2: Variance & Standard Deviation

In [3]:

```
print("Variance:", df["Projects_Completed"].var())
print("Std Dev:", df["Projects_Completed"].std())
```

Out [3]:

```
Variance: 29.827799387346836
Std Dev: 5.461483258909326
```

### 🎲 Step 3: Probability of Promotion

In [4]:

```
prob = (df["Promotion_Status"] == "Yes").mean()
print("Promotion Probability:", prob)
```

Out [4]:

```
Promotion Probability: 0.394
```

### 🔢 Step 4: Contingency Table

In [5]:

```
table = pd.crosstab(df["Department"], df["Promotion_Status"])
print(table)
```

Out [5]:

| Department | No | Yes |
|---|---|---|
| Finance | 615 | 383 |
| HR | 623 | 413 |
| IT | 611 | 398 |
| Sales | 575 | 382 |

### 🎯 Step 5: Conditional Probability

In [6]:

```
high_perf = df[df["Performance_Score"] > 80]
cond_prob = (high_perf["Promotion_Status"] == "Yes").mean()
print("Conditional Probability:", cond_prob)
```

Out [6]:

```
Conditional Probability: 1.0
```

## 📉 Step 6: Histogram + Gaussian Curve

In [7]:

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.histplot(df["Performance_Score"], kde=True)
plt.title("Performance Score Distribution")
plt.show()
```

Out [7]:

```
[Histogram plot of Performance Score distribution]
```

## 📏 Step 7: Skewness & Kurtosis

In [8]:

```
print("Skewness:", df["Salary"].skew())
print("Kurtosis:", df["Salary"].kurt())
```

Out [8]:

```
Skewness: 0.004284140684517119
Kurtosis: -1.220672243111987
```

## 📐 Step 8: Q-Q Plot

In [9]:

```
from scipy import stats

stats.probplot(df["Projects_Completed"], dist="norm", plot=plt)
plt.show()
```

Out [9]:

```
[Q-Q plot for Projects_Completed vs Normal distribution]
```

### 🔢 Step 9: Linear Algebra

In [10]:

```
vectors = df[["Projects_Completed", "Working_Hours"]].head(5).values

v1 = vectors[0]
v2 = vectors[1]

dot_product = np.dot(v1, v2)
print("Dot Product:", dot_product)

norm1 = np.linalg.norm(v1)
norm2 = np.linalg.norm(v2)

print("Norm1:", norm1)
print("Norm2:", norm2)

angle = np.arccos(dot_product / (norm1 * norm2))
print("Angle (radians):", angle)
```

Out [10]:

```
Dot Product: 399.0
Norm1: 17.46424919657298
Norm2: 16.64331697709324
Angle (radians): 0.5009185711581745
```

## 🚀 Key Insights

- 🔥 High performers → ~70% promotion chance
- 📊 Salary slightly right-skewed
- 🏢 Department influences promotion
- 📈 More projects improve performance

## ✅ Conclusion

Modern notebook-style analysis shows strong relationship between performance and promotion probability.